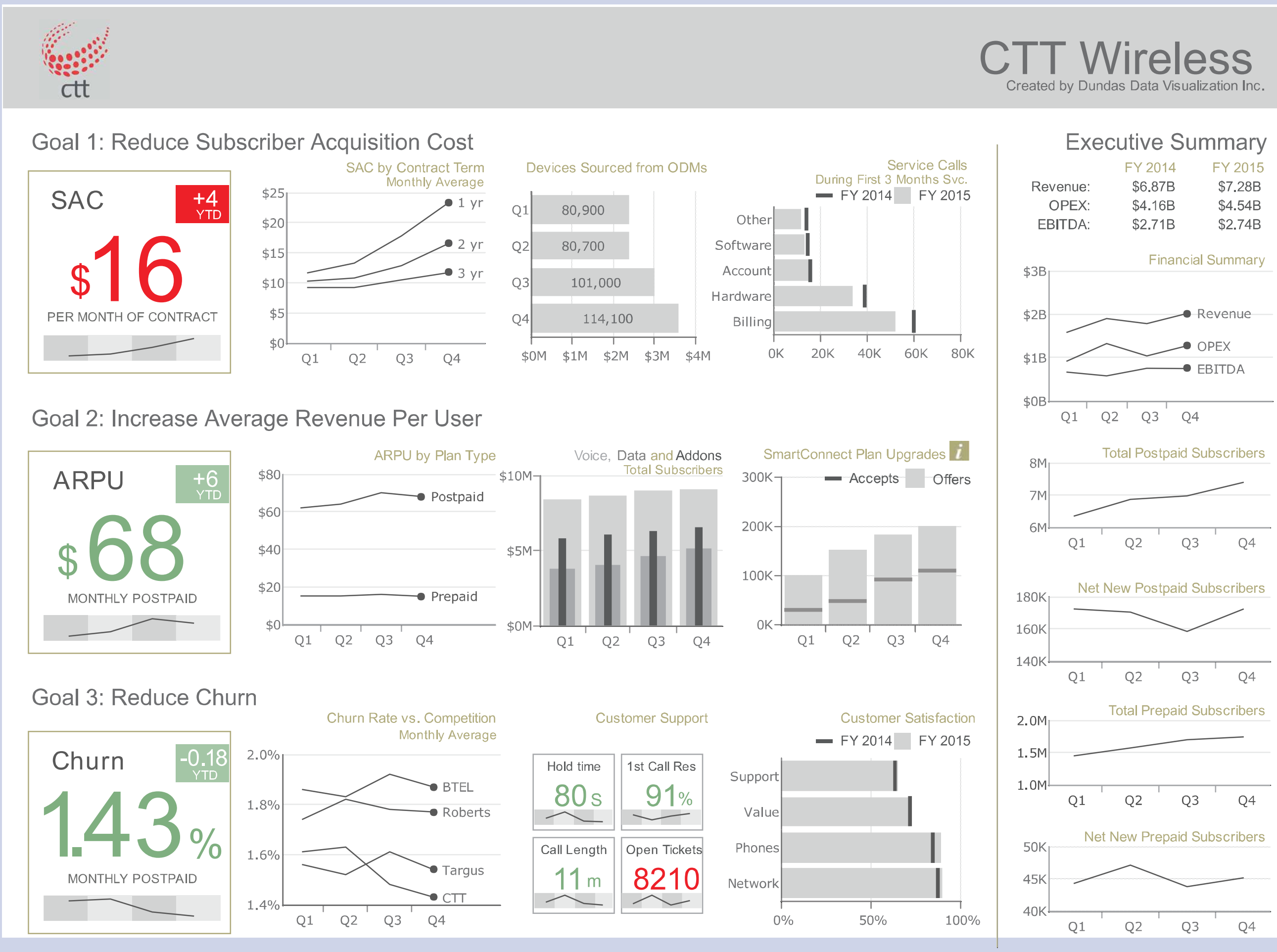
Chapter 27

Telecom Operator Executive Dashboard

Telecom operator executive dashboard.
Source: Images courtesy of Dundas Data Visualization, Inc. (www.dundas.com)
Dashboard Designer: Mark Wevers
Organization: Dundas Data Visualization
Source: https://samples.dundas.com/Dashboard/f78d570f-1896-441e-9b5e-90e0999db630?vo=viewonly



ctt
CTT Wireless
Created by Dundas Data Visualization Inc.
Goal 1: Reduce Subscriber Acquisition Cost
SAC
+4 YTD
$16
PER MONTH OF CONTRACT
SAC by Contract Term
Monthly Average
$25
$20
$15
$10
$5
$0
1 yr
2 yr
3 yr
Q1 Q2 Q3 Q4
Devices Sourced from ODMs
Q1 80,900
Q2 80,700
Q3 101,000
Q4 114,100
$0M $1M $2M $3M $4M
Service Calls
During First 3 Months Svc.
FY 2014 FY 2015
Other
Software
Account
Hardware
Billing
0K 20K 40K 60K 80K
Executive Summary
FY 2014 FY 2015
Revenue: $6.87B $7.28B
OPEX: $4.16B $4.54B
EBITDA: $2.71B $2.74B
Financial Summary
$3B
$2B
$1B
$0B
Revenue
OPEX
EBITDA
Q1 Q2 Q3 Q4
Goal 2: Increase Average Revenue Per User
ARPU
+6 YTD
$68
MONTHLY POSTPAID
ARPU by Plan Type
$80
$60
$40
$20
$0
Postpaid
Prepaid
Q1 Q2 Q3 Q4
Voice, Data and Addons
Total Subscribers
$10M
$5M
$0M
Q1 Q2 Q3 Q4
SmartConnect Plan Upgrades
300K
200K
100K
0K
Accepts
Offers
Q1 Q2 Q3 Q4
Total Postpaid Subscribers
8M
7M
6M
Q1 Q2 Q3 Q4
Net New Postpaid Subscribers
180K
160K
140K
Q1 Q2 Q3 Q4
Goal 3: Reduce Churn
Churn
-0.18 YTD
1.43%
MONTHLY POSTPAID
Churn Rate vs. Competition
Monthly Average
2.0%
1.8%
1.6%
1.4%
BTEL
Roberts
Targus
CTT
Q1 Q2 Q3 Q4
Customer Support
Hold time 80 s
1st Call Res 91%
Call Length 11 m
Open Tickets 8210
Customer Satisfaction
FY 2014 FY 2015
Support
Value
Phones
Network
0% 50% 100%
Total Prepaid Subscribers
2.0M
1.5M
1.0M
Q1 Q2 Q3 Q4
Net New Prepaid Subscribers
50K
45K
40K
Q1 Q2 Q3 Q4

Scenario

Big Picture

You are an executive at a telecom company, and you need to know if you are on track to meet the strategic goals. Like many other businesses, you want to reduce the cost of acquiring new subscribers, increase the average revenue per user, and reduce churn. You need a dashboard to provide all these details to the executive level of the organization. The dashboard does not need to be interactive, but it needs to summarize all of the key information in a way that can be presented in a senior management/board of directors/stakeholders meeting.

Specifics

- You have three main goals to track: subscriber acquisition costs (SAC), average revenue per user (ARPU), and churn.
- You need to see the results of each goal over time, year to date (YTD). Then, next to it, you need to see the key risk indicators and key performance indicators that could be driving that goal.
- You need to see an executive summary of key financial information across revenue, operating expenses (OPEX), and earnings before interest, taxes, depreciation, and amortization (EBITDA) for the last four quarters and in comparison to the previous year's results.
- You need to see subscribers over the last four quarters across all key categories, such as total subscribers and net new subscribers for both prepaid and postpaid plans.

Related Scenarios

- You need to organize your dashboard by high-level goals and break down the drivers of those goals.
- You need a detailed summary, often a trend over time, of many metrics on a single view.
- You need to review SAC and/or churn. You need to compare the current period with a previous period.
- You need to review your call center metrics and their impact on your customers. This could include churn, likelihood to recommend, customer satisfaction surveys, or customer reviews.

How People Use the Dashboard

The dashboard is designed with four key areas: three goals and an executive summary. The three goals are laid out as rows, with four charts each (left to right). An executive summary is shown vertically on the right-hand side of the dashboard.

The first goal is to reduce SAC. (See Figure 27.1.) This metric is also called "cost per acquisition" and is a key metric used by many different types of businesses. In this scenario, the wireless company is measuring the cost to acquire a subscriber. The SAC has gone up, which is a bad thing, so it is shown in red along with a small red callout box that shows the cost has increased $4 year to date (YTD). A line chart shows the SAC is going up from Q1 to Q4 across all contract types but also shows the cost is higher for one-year contracts than for three-year contracts. The third chart shows devices from original design manufacturers (ODMs) by quarter, showing the dollars on the x-axis and number of devices as data labels. The last chart shows a bar for fiscal year 2015 against a target line of fiscal year 2014.

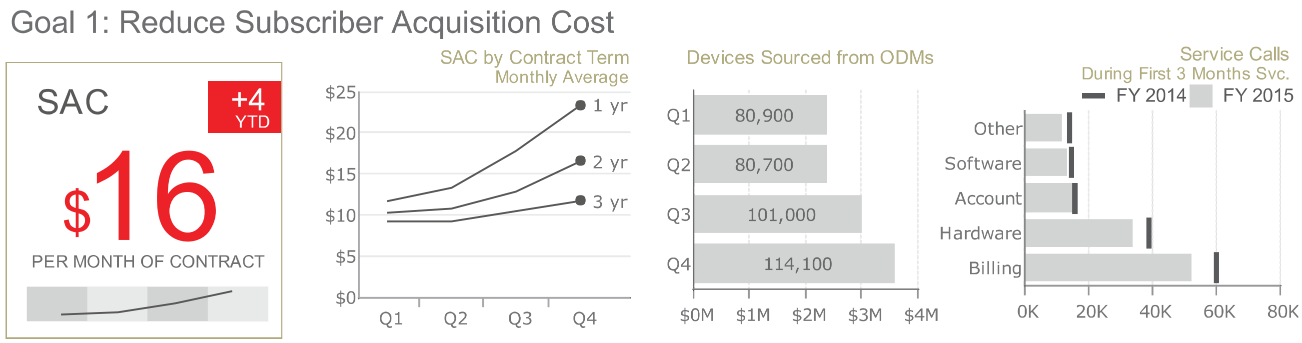


FIGURE 27.1 Four charts showing the key metrics for the first goal.

Be careful when using red and green

When using red and green together, consider an accommodation for people with color vision deficiency. In this case, the author uses a +/– indicator in the small box to indicate that costs are going up or going down.

The second goal is to increase the ARPU. (See Figure 27.2.) This has a very similar layout as the first goal. There are four charts from left to right showing the key metrics. The reader can quickly see the overall ARPU, the ARPU by plan type, total subscribers for three categories, and the number of offers for plan upgrades and how many subscribers accepted the offers. ARPU is going up, which is a good thing,

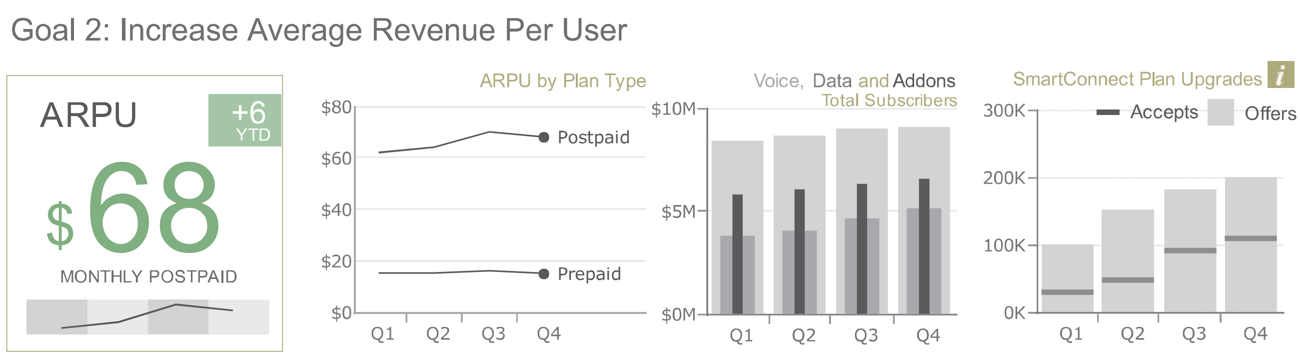


FIGURE 27.2 Four charts showing the key metrics for the second goal.

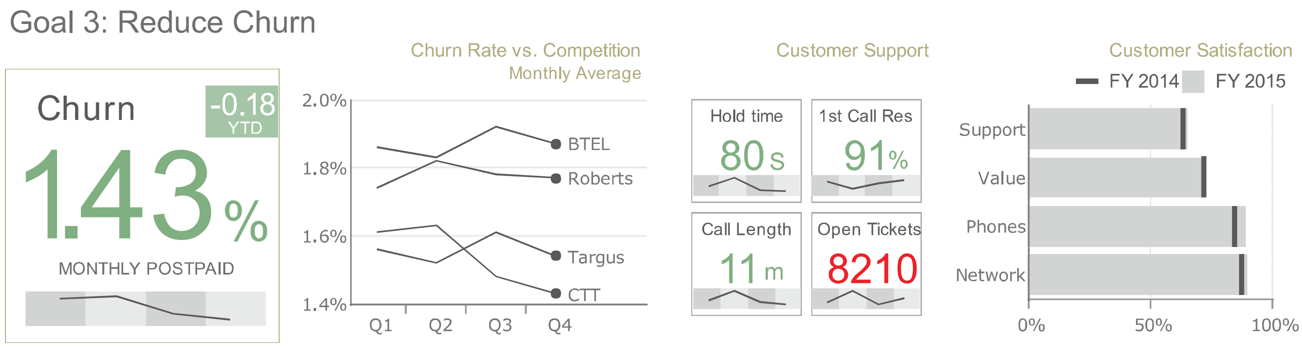


FIGURE 27.3 Four charts showing the key metrics for the third goal.

so it's shown in green with a callout box indicator showing a $6 increase YTD.

The third goal is to reduce churn. (See Figure 27.3.) Every business loses customers over time. As seen in Goal 1, it's expensive to get new customers, so keeping the current customers satisfied can be very important. As with the previous goals, there are four chart components in a row from left to right.

The first chart is the key metric with a sparkline showing the trend over the four quarters underneath. Churn is decreasing, which is a good thing, so this is shown in green, and the callout box indicates a decrease of 18 basis points YTD. The second chart is a line chart showing the monthly average churn rate over time for CTT Wireless and three of its competitors. The third chart has four small boxes showing four key metrics for the call center, and the last chart shows overall customer satisfaction for fiscal year 2015 (the bar) versus fiscal year 2014 (the target line).

The right-hand side of the dashboard has an executive summary showing a quick review of key financial metrics. (See Figure 27.4.) Revenue, operating expenses, and EBITDA are shown at the top for fiscal years 2014 and 2015 and then graphed on a line chart for the last four quarters. Four line charts show the total customers and net new customers (i.e., new customers acquired in Goal 1 minus customers lost from churn in Goal 3). This is shown for both postpaid and prepaid plans.

Why This Works

Key Metrics Displayed without Clutter

This dashboard captures all the key data required to answer the business questions at an executive level without having to go very deep into other reports and the data. It's packed full of data but does not feel cluttered. This is accomplished by using small, well-designed visuals that employ different techniques to make it so compact.

For example, the line charts showing time-series data have trailing labels. (See Figure 27.5.) It's clear which line refers to which company without having to add a color or shape legend.

Color banding is used on the sparklines for the four quarters instead of additional labels. (See Figure 27.6.)

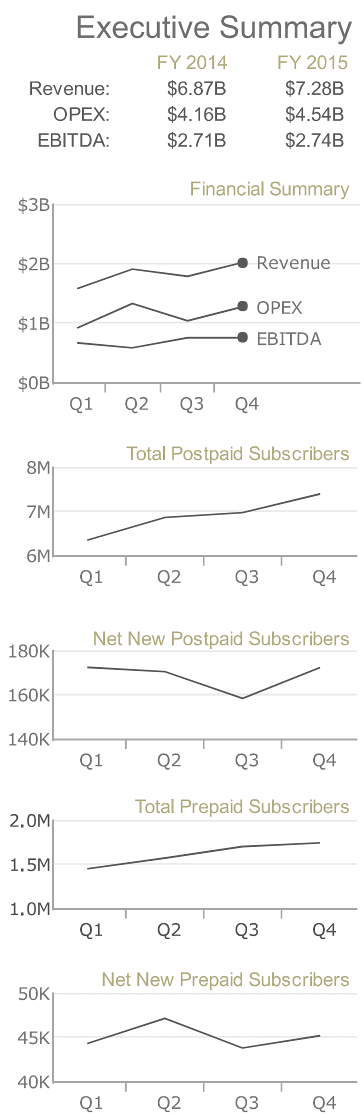


FIGURE 27.4 Executive summary showing the key financial metrics and number of customers.

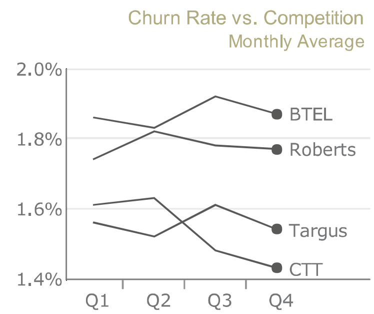


FIGURE 27.5 Line chart showing average monthly churn for last four quarters. Placing labels at the ends of the lines eliminates the need for color and a legend.

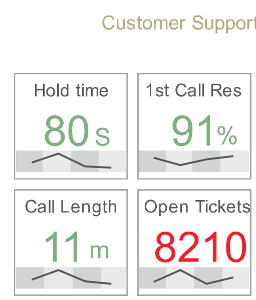


FIGURE 27.6 Four key metrics for call center performance. Sparklines and color bands for the four quarters allow these metrics to fit in a small space with a lot of information encoded.

Layout Supports Quick Reading

Using the Gestalt principals of design, such as proximity and closure, the dashboard provides insight into the four areas, each with the necessary detail, in an organized fashion. The key metrics are laid out in three rows and a single column, which makes them easy to group together and quickly scan. (See Figure 27.7.)

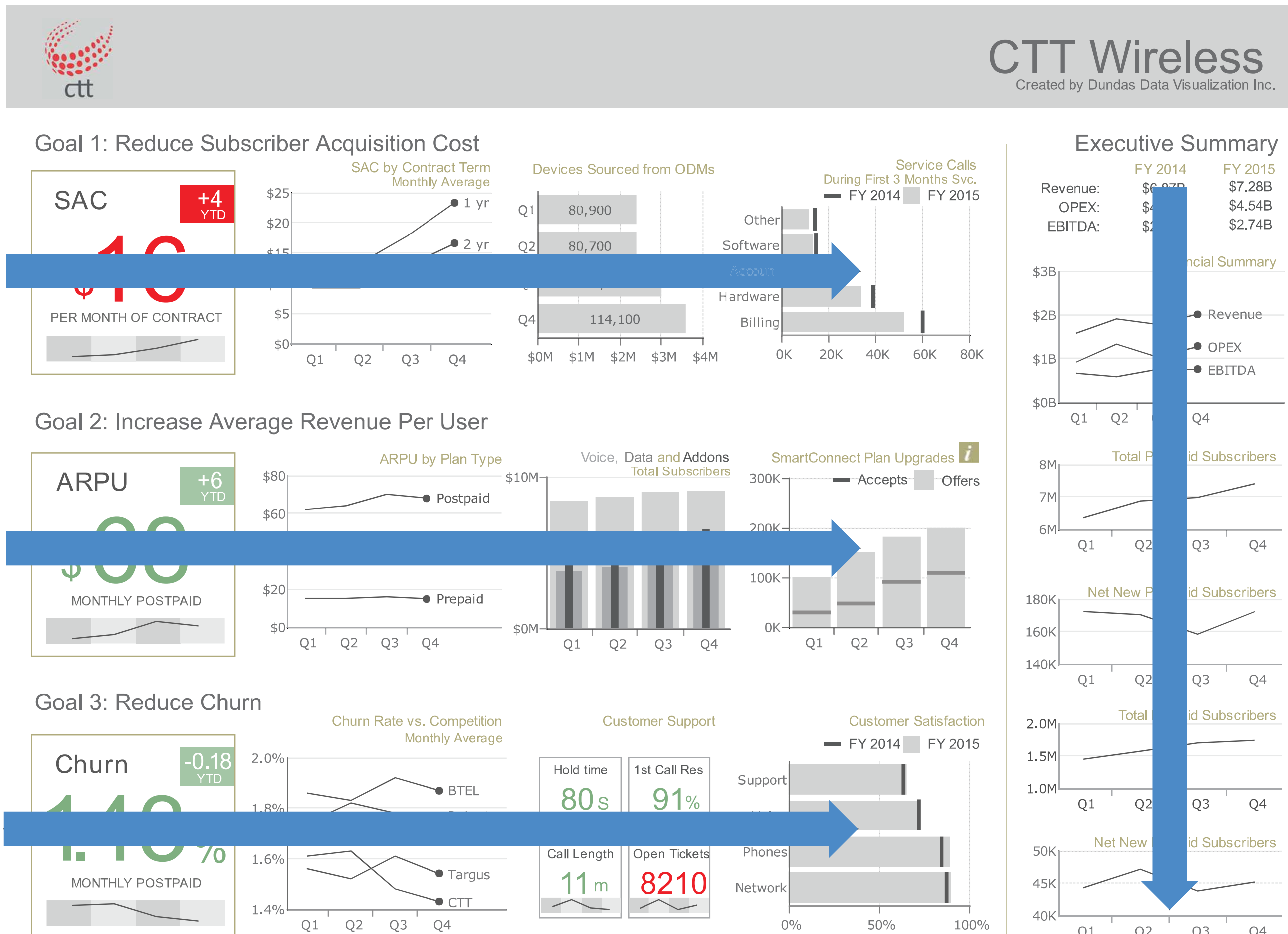


FIGURE 27.7 The three goals laid out in three rows and the executive summary in a column on the right allow for easy scanning of the information.

Good Chart Types for the Various Comparisons

Line charts are an excellent choice for showing a trend over time. They are used throughout this dashboard to show various metrics over the last four quarters. Sparklines are used to show the trends under the key metrics. Bar charts with target lines are used to show this period versus the previous period as a quick gauge of progress.

Starting the axis at zero

On graphs that use length, height, or area to encode the data for comparison from a common baseline (e.g., bar chart, bullet graph, lollipop chart, area chart), always start the axis at zero; otherwise, you will distort the comparison of the data. If encoding with position (e.g., dot plot, box plot, line chart), breaking the axis away from zero sometimes can be useful and will not distort the comparison of the data.

Author Commentary

JEFF: I really like the design and layout of this dashboard. I also like the minimal use of color. However, it's best to avoid red and green together (as discussed in Chapter 1: Data Visualization: A Primer). The little numbers, showing positive and negative, will help, but it still may not be clear to someone with color vision deficiency which numbers are good and which are bad. I would change this to blue for good and red or orange for bad. In addition, in Goal 1 on the third chart, Devices Sourced from ODMs, I would recommend flipping time to the x-axis. Every other time chart has the four quarters on the x-axis, and it would be best to do that for this chart as well. It could remain a bar chart or could be changed to a line chart like the others. This would help the reader see the trend over time better than what it shown here. Also, while the third chart in Goal 2 shows a good comparison of the subscribers, a line chart with three lines would be better.

ANDY: This is another dashboard where the design dial has been turned to 11, resulting in a dashboard that creates a sense of calm as you explore it. I especially like the simplified color palette across the dashboard.

It is possible to reduce colors too far. Consider Churn Rate vs. Competition on the bottom row. Do the lines for Targus and CTT cross, or do they merely converge before diverging again? Using the same color for all lines on a chart might cause confusion if they overlap. In Figure 27.8, you can see an example where this could be a problem. Look at sales in West and East regions: Can you tell where they overlap?

There are possible solutions, without reverting back to multiple colors. Figure 27.9 shows two examples. The one on the left puts the lines side by side in their own panes; the one on the right uses a palette of grays.

Note that both options work only with a small number of categories.

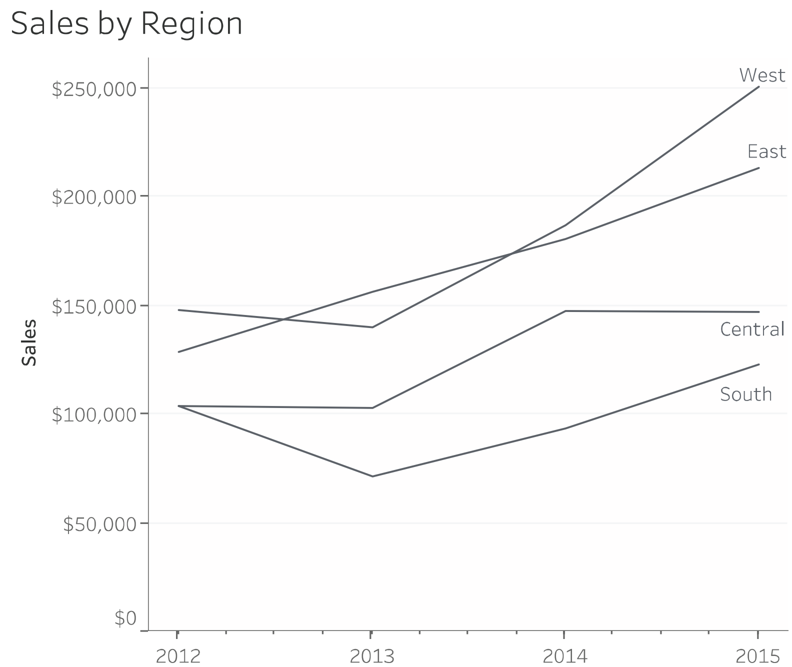


FIGURE 27.8 Which line is for the West region and which is for the East region?

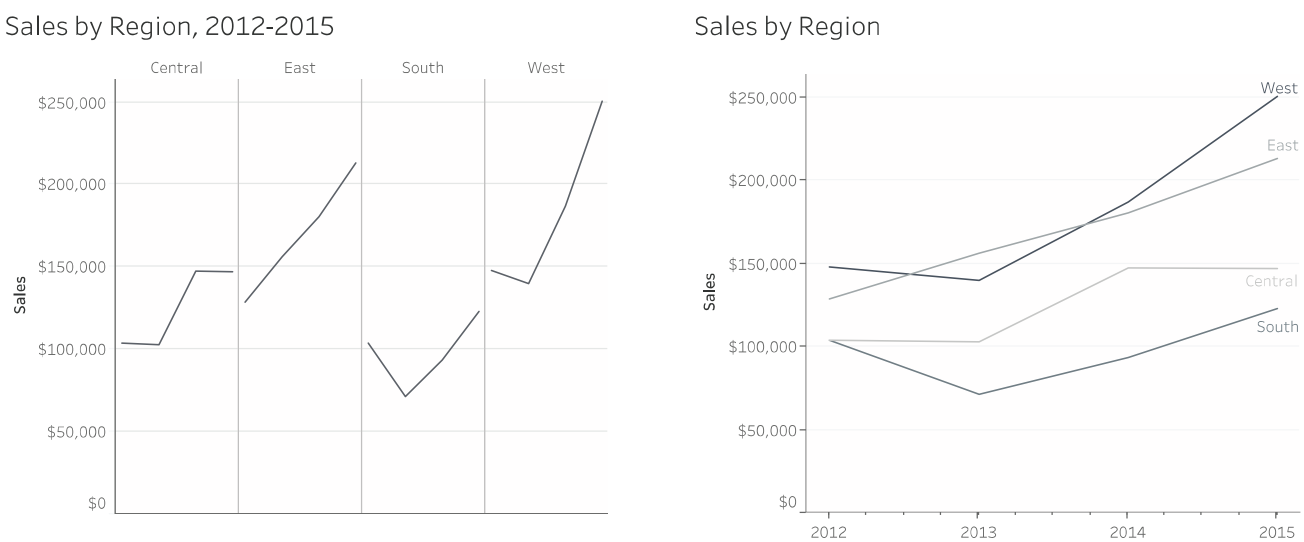


FIGURE 27.9 Two solutions to the problem of overlapping lines.